# Temujin Orkhon

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## Education

**University of California, Berkeley** Berkeley, CA
PhD Candidate in Chemistry Expected June 2031
Concentration: Quantum Chemistry, Computational Chemistry, Quantum Computing

**Massachusetts Institute of Technology** Cambridge, MA
B.S. in Chemistry & Biology, B.S. in Physics, Minor in Computer Science June 2026

## Experience

**Learning Matters Lab (Gómez-Bombarelli Group)** Cambridge, MA
Undergraduate Researcher January 2023- June 2024

- Constructed multi-layered machine learning models, utilizing uncertainty as a learnable parameter, achieving greater generalizability and accuracy in predicting molecular property
- Developed software aimed at generating artificial datasets, implementing, and benchmarking existing molecular property machine learning models

**Elkin Lab** Cambridge, MA
Undergraduate Researcher January 2025- June 2025

- Designed novel algorithms to extract organic transformation templates, improving on literature template extractors through the usage of context-aware reaction centers
- Helped in the development of software that streamlines the analysis of total synthesis data to determine impactful multistep transformations for further experimental optimization

**Van Voorhis Group** Cambridge, MA
Undergraduate Researcher June 2024- June 2026

- Developed electronic structure methods employing quantum embedding to calculate the electronic properties of strongly correlated chemical systems.
- Simulated and analyzing defected semiconductor materials for applications in quantum computing, collaborating with outside universities

## Publications & Presentations

- **Orkhon T.**; Weatherly S.; Van Voorhis T. (2025), "Extracting band structures using Bootstrap Embedding
- **Contributor** (2025), "QuEmb: A Toolbox for Bootstrap Embedding Calculations of Molecular and Periodic Systems", Journal of Physical Chemistry, doi.org/10.1021/acs.jpca.5c02983
  Contributed by testing the periodic bootstrap embedding code, and helping with bug fixes
- **Orkhon T.**; Rein J.; Elkin M. (2025), "Holistic reaction template generation using a context-driven approach" Presented at the Chem UROP Symposium, Cambridge, MA
- **Orkhon T.**; Greenman K.; Green W.; Gómez-Bombarelli R. (2024) "Uncertainty improves multi-fidelity machine learning", Presented at the ChemE UROP Symposium, Cambridge, MA
- Greenman K.; **Orkhon T.**; Green W.; Gómez-Bombarelli R. (2023), "Multi-fidelity deep learning for data-efficient molecular property models from experimental and computational data", AIChE Annual Meeting

## Teaching Experience

**MIT Department of Chemistry** Cambridge, MA
Chemistry Tutor February 2024- June 2026

- Provided one-on-one tutoring in general chemistry, organic chemistry, and physical chemistry to MIT students interested in chemistry.
- Created custom lesson plans, practice problems, and review materials tailored to the student's needs.

**Talented Scholars Resource Room** Cambridge, MA
Facilitator Sept. 2023- June 2024

- Facilitated student groups, providing comprehensive assistance with problem sets and hosting exam reviews, improving academic performance of students at MIT.
- Delivered personalized one-on-one guidance to MIT students in fundamental introductory subjects, ensuring an effective approach.

**TomYo EdTech** Ulaanbaatar, Mongolia
College Counselor August 2023- Sept. 2024

- Advocated to Mongolian students to study abroad, guiding high school students through the college application process
- Counseled prospective students on their higher education plans, offering valuable insights and guidance to align their goals with suitable academic pathways.

## Honors & Awards

- Sigma Pi Sigma (Physics Honor Society)
- Academic achievement award (MIT Chemistry 2026)
- Miyagawa Prize winner (MIT Global Languages 2026)
- Best poster presentation in the ChemE UROP Symposium (2024)
- International Chemistry Olympiad Silver medalist (2021, 2022)
- International Mendeleev Chemistry Olympiad Silver medalist (2021)

## Technical & Laboratory Skills

**Languages:** English, Mongolian, Japanese
**Computational:** Python, Pandas, Numpy, PyTorch, SciPy, FHI-aims, PySCF, QuEmb, QChem
**Laboratory**: Titration, Distillation, Thin-Layer Chromatography, Column Chromatography, UV-vis Spectroscopy, Fluorescence Spectroscopy, FTIR Spectroscopy
**Interests/Hobbies:** Photography, Football (Soccer), Basketball